CS683 Programming Assignment II
Data Prefetching and Cache Replacement

Deepjyoti Kakoti (26M0774)
Debasish Bordoloi (26M0776)
Rahul Paul (26M2102)
Piyush Malkhede (26M2130)

15th October, 2026

Summary of results

Task	Mechanism (baseline)	Avg. speedup	Best trace
1B	SPP at L1D (no PF)	1.0024	Trace1: 1.0042
2A	I-RRIP at L2 (LRU)	1.0079	Trace2: 1.0241
2B	Dynamic I-RRIP at L2 (LRU)	1.0068	Trace2: 1.0223
3	SPP + PF-aware Dyn-I-RRIP (no PF)	1.0069	Trace2: 1.0163

Table 1: Headline results (baseline in parentheses; PF = L1D prefetching). Average speedups are geometric means over the three traces.

- **Task 1 (prefetching):** SPP combines a uniform-stride predictor with a signature-based pattern table and accuracy-based throttling. It is more accurate than IP-stride on every trace and has the lowest L1D load MPKI, but both prefetchers give speedups below 0.6% because at most 5.8% of the L1D load misses are predictable from per-IP strides. Storage: about 6.7 KB.
- **Task 2A (I-RRIP):** protecting instruction lines at insertion, promotion and eviction cuts L2 instruction misses on every trace and gives 2.4% speedup on the code-heavy Trace2. Storage: 3 KB.
- **Task 2B (Dynamic I-RRIP):** set dueling with 32 SRRIP leader sets keeps about 87% of the I-RRIP gain while being able to fall back to SRRIP. The result is insensitive to the threshold and the number of leaders. Storage: 3 KB + 60 bits.
- **Task 3 (SPP + prefetch-aware Dynamic I-RRIP):** prefetched lines are inserted into the L2 at RRPV 3, prefetch hits are not promoted and prefetches are not counted by the leader sets. The combination keeps SPP's L1D gains and most of Dynamic I-RRIP's L2 instruction-miss reduction, and has the highest average IPC, but its speedup only matches Dynamic I-RRIP's and it does not beat both standalone mechanisms on every trace. No extra storage.

Experimental setup

All runs use the ChampSim build from the PA2 repository with 25M warmup and 25M simulated instructions on the three provided traces (Trace1-003, Trace2-001, Trace3-002). Each experiment changes only the mechanism under study:

- **Task 1**: only the L1D prefetcher changes; every cache uses LRU and all other prefetchers are off. Baseline: no L1D prefetching.
- **Task 2**: only the L2 replacement policy changes; all prefetchers are off and every other cache uses LRU. Baseline: LRU at the L2 (1024 sets $\times$ 8 ways, 512 KB).
- **Task 3**: SPP at the L1D together with prefetch-aware Dynamic I-RRIP at the L2; every other cache uses LRU. Baseline: no L1D prefetching with LRU at the L2 (the Task 1B baseline).

Metrics:

- **Speedup** = IPC with the mechanism/IPC of the baseline on the same trace. The "Average" speedup is the geometric mean over the three traces; every other average is the arithmetic mean.
- **MPKI**: misses per 1000 instructions, from the ChampSim statistics (`L1D LOAD` line for Task 1; `L2C LOAD`, `L2C INSTRUCTION LOAD` and `L2C DATA LOAD` lines for Task 2).
- **Prefetch accuracy**: the `ACCURACY` field of the `L1D USEFUL LOAD PREFETCHES` line (useful load prefetches / prefetches issued to the lower level).
- **Prefetch coverage** $= \dfrac{\text{L1D load misses (baseline)} - \text{L1D load misses (prefetcher)}}{\text{L1D load misses (baseline)}}$.

The raw outputs are in `results/task1b`, `results/task2a`, `results/task2b` and `results/task3`. Task 1 tables and plots were produced by `scripts/task1b_metrics.py`; Task 2 tables and plots by `scripts/parse_results.py`, `scripts/plot_results.py` and `scripts/plot_task2b.py`; Task 3 tables and plots by `scripts/task3_metrics.py`.

1 Task 1A: IP-stride prefetcher

`prefetcher/ip_stride.l1d_pref` keeps a 24-entry, fully associative table of *IP trackers*. Each tracker holds the low 8 bits of the load IP (`MASK_IP = 0xFF`), the last cache-line address that IP touched, the last stride it saw and an LRU rank.

On every L1D access the prefetcher:

1. Looks up the IP. On a miss it replaces the LRU tracker, records the current line address with stride 0 and returns without prefetching.
2. On a hit it computes the stride (in cache lines) from the last address. A zero stride is ignored.
3. If the new stride equals the stored stride (the same stride seen twice in a row), it issues prefetches for addr $+ k \cdot$ stride, $k = 1 \ldots 3$ (`PREFETCH_DEGREE = 3`). It stops at the first target that leaves the 4 KB page of the demand access. A prefetch fills the L1D when fewer than half of the MSHRs are in use and only the L2 otherwise.
4. It stores the new address and stride and moves the tracker to MRU.

The design has no confidence counter (a single repeat of a stride is enough to prefetch), a fixed degree and no feedback on whether its prefetches are useful. It only captures constant strides: a repeating multi-stride sequence such as $(+1, +2, +2, +2, +3)^n$ almost never shows two equal strides in a row in a way that predicts the next one. SPP (Task 1B) addresses all three points.

2 Task 1B: Stride Pattern Prefetcher (SPP)

2.1 Design

SPP (`prefetcher/spp.l1d_pref`) works on cache-line addresses and keeps one table indexed by the low 8 bits of the IP. Each IP entry contains two predictors that are trained on every access with a non-zero stride:

Uniform-stride predictor. Each IP entry has a *monitored stride* and a 2-bit saturating confidence counter (0 = no confidence, 2 = high confidence). If the new stride equals the monitored stride the counter is incremented; otherwise it is decremented, and once it is already 0 the monitored stride is replaced by the new stride. So a constant stride needs to repeat twice before it is trusted, and a broken pattern loses the right to prefetch after a single mismatch.

Non-uniform-stride predictor. Each IP entry keeps the history of its last 8 strides (`HISTORY_LEN = 8`). This history is the *signature*. A per-IP pattern table of up to 24 entries maps a signature to the stride that followed it, with a 7-bit confidence and an LRU age. On every access the entry for the *previous* signature is trained with the new stride: a correct prediction increments its confidence, a wrong prediction halves it, and a wrong prediction at confidence 0 replaces the predicted stride. The new stride is then shifted into the history. When the pattern table is full, the entry with the lowest confidence − age is evicted, so stale and unreliable patterns go first. An 8-stride signature is long enough to tell apart the positions inside a repeating sequence such as $(+1, +2, +2, +2, +3)^n$, where the `+2` stride is followed by different strides depending on where it occurs.

Classification and prefetching. The uniform predictor has priority. If the IP's uniform confidence is high (2), SPP prefetches line $+ k \cdot$ stride for $k = 1 \ldots$ degree. Otherwise it falls back to the pattern predictor: it looks up the current signature, takes the predicted stride, shifts it into a copy of the signature and repeats the lookup, walking up to *degree* steps ahead as long as each signature is found. An IP therefore moves to non-uniform prediction as soon as its constant stride breaks, and back to uniform prediction once the constant stride has repeated twice.

Page boundary and MSHR. No prefetch crosses the 4 KB page of the demand access: the uniform path stops at the first out-of-page target, and the pattern path skips out-of-page targets. Prefetches are only issued while at least 3 MSHR entries are free, so they never starve demand misses. Strides outside $[-128, 127]$ lines cannot be stored in 8 bits; such an access counts as a uniform-stride mismatch, clears the stride history and issues no prefetch.

Accuracy-based throttling. The prefetch degree is global and starts at 1. Every 256 trained accesses (`EPOCH_LEN`) SPP reads ChampSim's L1D counters for prefetches issued to the lower level

and useful prefetches, and computes the accuracy of the prefetches issued during that epoch:

$$\text{acc} = \frac{\Delta\,\text{useful}}{\Delta\,\text{issued}}, \qquad \text{degree} \leftarrow \begin{cases} \min(\text{degree}+1,\ 8) & \text{acc} > 0.6 \\ \max(\text{degree}-1,\ 1) & \text{acc} < 0.3 \\ \text{degree} & \text{otherwise} \end{cases}$$

Epochs in which no prefetch was issued leave the degree unchanged.

2.2 Results

	IPC			Speedup		L1D load MPKI		
Trace	None	IP-stride	SPP	IP-stride	SPP	None	IP-stride	SPP
Trace1-003	0.2322	0.2334	0.2332	1.0051	1.0042	10.679	10.119	10.062
Trace2-001	0.2511	0.2512	0.2513	1.0002	1.0008	13.534	13.436	13.446
Trace3-002	0.3821	0.3830	0.3829	1.0024	1.0023	3.403	3.318	3.285
Average	0.2885	0.2892	0.2891	1.0026	1.0024	9.205	8.957	8.931

Table 2: IPC, speedup over no prefetching and L1D load MPKI.

	Accuracy (%)		Coverage (%)		Prefetches issued		Useful prefetches	
Trace	IP-stride	SPP	IP-stride	SPP	IP-stride	SPP	IP-stride	SPP
Trace1-003	52.79	60.72	5.24	5.78	30988	28454	16359	17278
Trace2-001	38.13	57.47	0.73	0.65	11472	5211	4374	2995
Trace3-002	16.09	33.36	2.51	3.48	17013	9496	2738	3168
Average	35.67	50.52	2.83	3.30				

Table 3: L1D prefetch accuracy, coverage and prefetch counts.

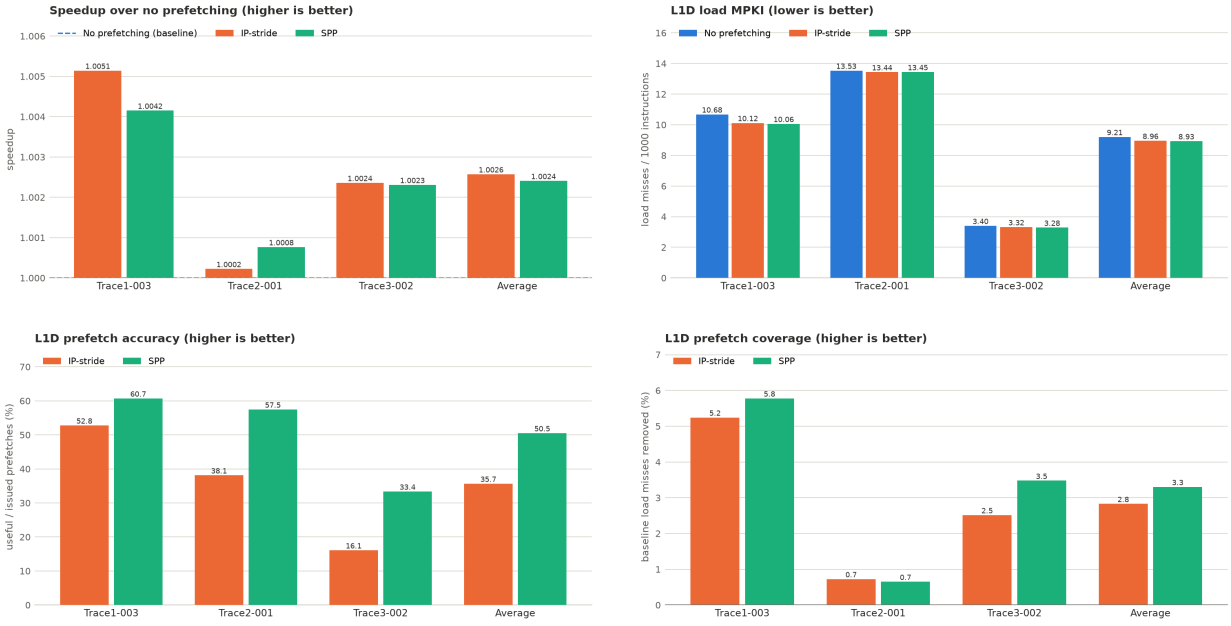


Figure 1: Speedup, L1D load MPKI, prefetch accuracy and coverage of IP-stride and SPP.

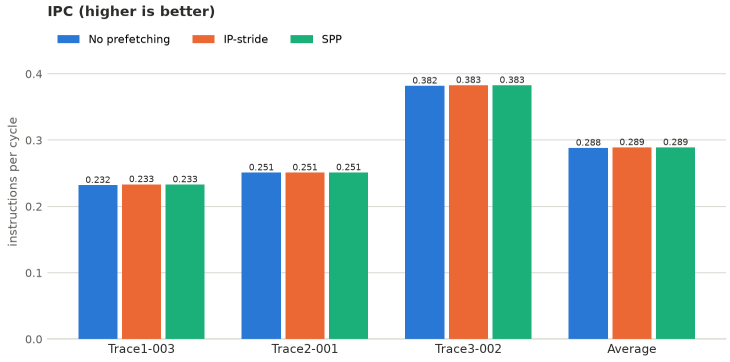


Figure 2: IPC with no prefetching, IP-stride and SPP.

Observations.

- **Accuracy.** SPP is more accurate than IP-stride on every trace (50.5% vs. 35.7% on average). On Trace2 and Trace3 it doubles or nearly doubles the accuracy while issuing about half as many prefetches (5211 vs. 11472 and 9496 vs. 17013). The confidence counters and the throttle stop SPP from prefetching on strides that do not repeat, which IP-stride does after a single repeat with a fixed degree of 3.
- **Coverage and MPKI.** SPP has the higher average coverage (3.30% vs. 2.83%) and the lowest average L1D load MPKI (8.931). On Trace1 and Trace3 it turns fewer prefetches into more useful ones (e.g. 3168 vs. 2738 useful on Trace3), which is where the pattern predictor adds coverage beyond constant strides. Only on Trace2 is its coverage slightly lower (0.65% vs. 0.73%), because the throttle keeps it at a low degree.

- **Speedup.** Both prefetchers give small speedups: 0.26% (IP-stride) and 0.24% (SPP) geometric mean. SPP is best on Trace2, equal on Trace3 and slightly behind on Trace1, despite its lower MPKI there. IP-stride issues more prefetches and sends them to the L2 when the L1D MSHRs are busy, so some of its benefit likely shows up as L2 hits that the L1D coverage metric does not count, whereas SPP drops prefetches when fewer than 3 MSHRs are free.

2.3 Storage overhead

Table 4 gives the bit-level breakdown of every table entry with the field widths used by the implementation. The IP tag is the key of the IP table and the signature is the key (tag) of the pattern table.

Structure	Field	Bits
IP entry	IP tag (low 8 bits of the IP)	8
	Last cache-line address	64
	Stride history / signature ($8 \times$ 8-bit strides)	64
	Monitored (uniform) stride	8
	Uniform-stride confidence	8
	LRU age	8
	Total per IP entry (without pattern table)	**160**
Pattern entry	Signature tag ($8 \times$ 8-bit strides)	64
	Next stride	8
	Confidence	8
	LRU age	8
	Total per pattern entry	**88**
Per IP	1 IP entry + 24 pattern entries ($160 + 24 \times 88$)	2272
Throttle (global)	Prefetch degree	32
	Epoch access counter	64
	Snapshot of prefetches issued	64
	Snapshot of useful prefetches	64
	Total	**224**
Total: $24 \times 2272 + 224$		**54752 bits $\approx$ 6.68 KB**

Table 4: SPP storage. 24 IP entries, each with a 24-entry pattern table. The issued/useful counters themselves are ChampSim's existing L1D statistics and are not counted.

Almost all of the storage (93%) is in the pattern tables. Several fields are wider than they need to be: the uniform confidence only takes the values 0–2 (2 bits), the pattern confidence never exceeds 127 (7 bits), a cache-line address of a 64-bit virtual address needs 58 bits, the degree (1–8) fits in 3 bits and the epoch counter (0–255) in 8 bits. With these widths the total drops to about 6.4 KB.

2.4 Choice of throttling thresholds and degrees

The thresholds and maximum degree were chosen with a sweep on Trace1-003, the trace with the most prefetch activity (Table 5, Figures 3 and 4). Each run varies the lower threshold (MIN_CONF:

below it the degree decreases), the upper threshold (MAX_CONF: above it the degree increases) or the maximum degree, and is compared against the no-prefetching baseline of Trace1-003.

MIN	MAX	Max deg.	IPC	Speedup	MPKI	Acc. (%)	Cov. (%)	PF issued
0.3	0.6	4	0.2328	1.0028	10.128	62.75	5.16	24247
0.4	0.7	4	0.2328	1.0026	10.159	63.22	4.87	22478
0.5	0.75	4	0.2328	1.0025	10.193	62.50	4.54	21207
0.6	0.85	4	0.2328	1.0025	10.223	62.44	4.26	19778
0.7	0.9	4	0.2327	1.0023	10.238	62.22	4.13	19167
0.5	0.75	2	0.2328	1.0024	10.248	61.86	4.03	18787
0.5	0.75	4	0.2328	1.0025	10.193	62.50	4.54	21207
0.5	0.75	6	0.2328	1.0027	10.159	63.02	4.86	22574
0.5	0.75	8	0.2328	1.0028	10.149	63.49	4.96	22846

Table 5: SPP throttle sweep on Trace1-003 (MIN/MAX = MIN_CONF/MAX_CONF, MPKI = L1D load MPKI). Top: threshold pairs at maximum degree 4. Bottom: maximum degree at thresholds 0.5/0.75.

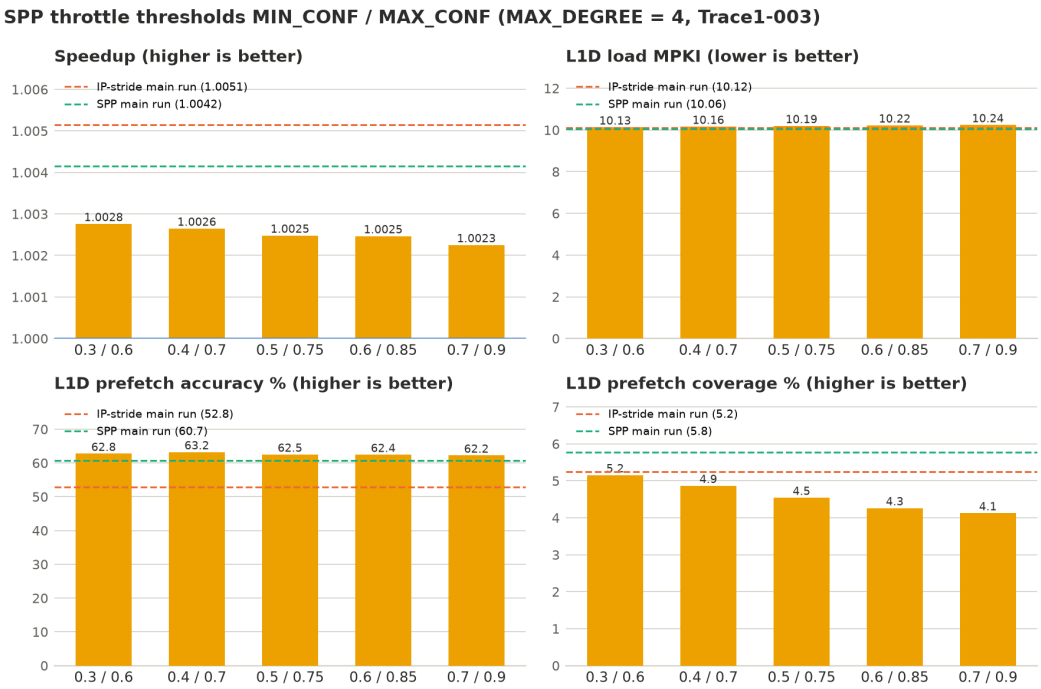


Figure 3: Sweep over the accuracy thresholds (maximum degree 4).

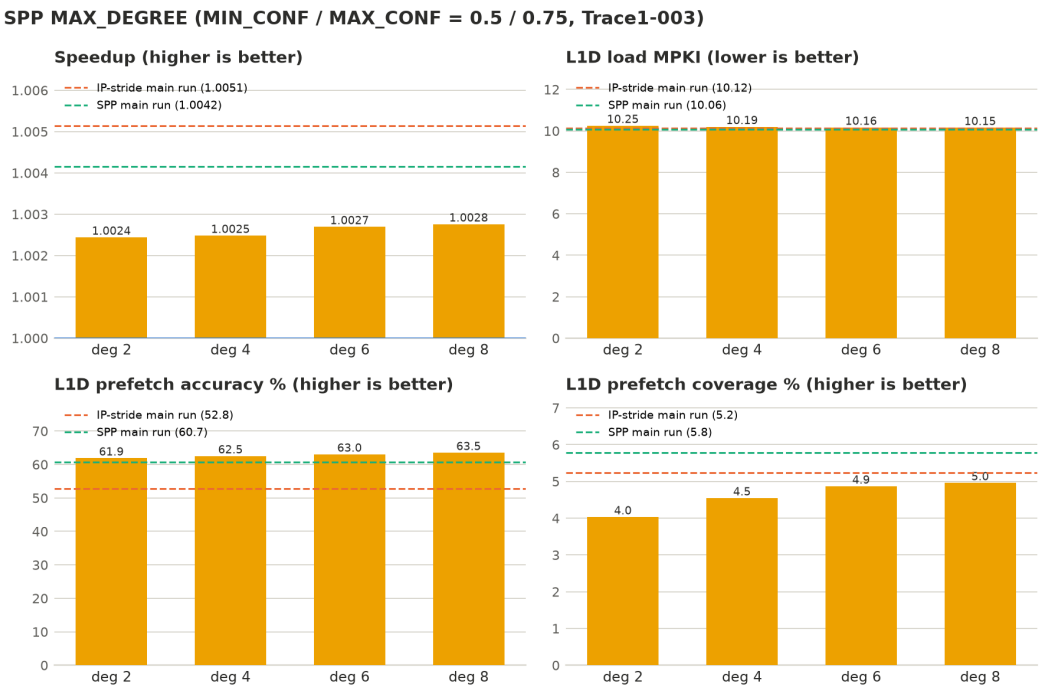


Figure 4: Sweep over the maximum degree (thresholds 0.5/0.75).

How the thresholds were decided. SPP's epoch accuracy on this trace sits around 62–63% for every setting. With high thresholds (e.g. 0.7/0.9) that accuracy is rarely above MAX_CONF, so the degree seldom grows, and it is often below MIN_CONF, so the degree is pulled down; SPP issues fewer prefetches (19167) and coverage falls to 4.13%. Lowering the thresholds increases the speedup monotonically, from 1.0023 at 0.7/0.9 to 1.0028 at 0.3/0.6, and coverage rises from 4.13% to 5.16%, while accuracy does not drop (62.2% → 62.8%). The extra prefetches are as useful as the rest, so being more aggressive does not cause pollution here. We therefore use **MIN_CONF = 0.3** and **MAX_CONF = 0.6**:

- Accuracy above 60%: more than every second prefetch is used, so prefetching further ahead is worth the bandwidth; the degree grows.
- Accuracy between 30% and 60%: the prefetches still help but are not reliable enough to issue more of them; the degree is kept.
- Accuracy below 30%: more than two out of three prefetches are wasted and only pollute the L1D and occupy MSHRs; the degree shrinks, down to 1.

We did not go below 0.3/0.6: the low threshold is the only protection against pollution on phases or traces where the patterns break down (Trace3 has an accuracy of only 33%), and on traces like that a lower threshold would leave SPP running at a high degree with mostly useless prefetches.

Why these degrees. The degree starts at 1, so an IP's first predictions are cheap, and changes by one step per epoch, so a single noisy epoch cannot swing it far. At thresholds 0.5/0.75 raising the maximum degree from 2 to 8 increases the speedup monotonically (1.0024 → 1.0028), coverage (4.03% → 4.96%) and even accuracy (61.9% → 63.5%): when patterns are confident, prefetching

further ahead hides more of the miss latency without polluting the cache. We use **a maximum degree of 8**. A larger maximum is of limited use because a prefetch never crosses the 4 KB page (64 lines): at strides of a few lines, 8 steps already reach much of the page, and the remaining targets would be dropped anyway. The minimum degree of 1 keeps SPP training and measuring its accuracy even in bad phases, so it can raise the degree again once its prefetches become useful.

The final configuration (0.3/0.6, maximum degree 8) combines the best setting from each sweep. With it, SPP reaches a speedup of 1.0042 on Trace1-003, higher than any setting in the sweep, with 60.7% accuracy and 5.78% coverage (Table 3). The sweep was run only on Trace1-003 and before the final runs, so its absolute numbers are best compared with each other rather than with the main results.

3 Task 2A: Instruction-Conscious L2 Replacement (I-RRIP)

Policy file: `replacement/irrip.l2c_repl`.

3.1 In short

- The L2 cache holds both program code (instruction lines) and data. When it is full, the replacement policy decides which line to evict.
- I-RRIP tries harder to keep instruction lines, because when code is missing the CPU cannot even fetch its next instruction.
- Result: fewer L2 instruction misses on every trace, and on the code-heavy trace (Trace2) the program runs 2.4% faster than with LRU. Average over the three traces: 0.79% faster.

3.2 Telling the policy "code or data?"

Originally the replacement policy had no way to know whether a line held code or data. We added one extra input, `isInstruction`, to `update_replacement_state()`, which the cache calls on every hit and every fill. It is 1 for an instruction line and 0 for everything else. Because all caches share this function's signature, the input was added everywhere it is declared, defined and called.

The value is computed by a small helper in `cache.cc`: a request is an instruction line only when it is a normal read (LOAD) or a PREFETCH and its `packet.instruction` flag is set. This is the same rule the simulator already uses to mark L2 lines as instruction lines, so address-translation lookups and writebacks are never mistaken for code.

3.3 The policy

I-RRIP starts from SRRIP (`rrip.l2c_repl`). Every line has a 2-bit re-reference prediction value (RRPV) from 0 to 3: 0 = "will be used again soon, keep me", 3 = "not needed soon, evict me first". I-RRIP changes how these values are assigned and used so that code lines stay longer (Table 6; the reasoning is in Section 3.6).

Part	SRRIP	I-RRIP
Insertion, instruction line	RRPV 2	RRPV 1
Insertion, data line	RRPV 2	RRPV 2
Promotion (hit), instruction line	RRPV 0	RRPV 0
Promotion (hit), data line	RRPV 0	RRPV 1
Promotion on a writeback hit	RRPV 0	no change
Eviction	first line at RRPV 3	first data line at RRPV 3, then any line at RRPV 3
Empty way in the set	not checked	used first

Table 6: SRRIP vs. I-RRIP rules. A hit never raises a line's RRPV.

3.4 Experimental setup

- 25M warmup + 25M measured instructions on each of the three traces.
- No L1D prefetcher; every other cache uses LRU. The baseline is LRU at the L2 (1024 sets × 8 ways, 512 KB).
- Speedup = IPC with the policy / IPC with LRU, on the same trace. The average is the geometric mean.
- MPKI = misses per 1000 instructions (lower is better). We report the L2C load MPKI and its two parts: instruction and data.
- The simulator is deterministic (fixed random seeds), so one run per setting is enough. Re-running from the final code gave identical results.

3.5 Results

Trace	Policy	IPC	Speedup	L2C load MPKI	L2 instr. MPKI	L2 data MPKI
Trace1	LRU (baseline)	0.2323	1.0000	17.696	10.338	7.358
	SRRIP	0.2322	0.9995	17.639	10.562	7.077
	I-RRIP	0.2322	0.9994	17.649	9.765	7.884
Trace2	LRU (baseline)	0.2512	1.0000	27.357	18.853	8.504
	SRRIP	0.2518	1.0026	25.746	17.781	7.965
	I-RRIP	0.2572	1.0241	21.329	12.204	9.125
Trace3	LRU (baseline)	0.3821	1.0000	3.954	1.124	2.830
	SRRIP	0.3817	0.9991	4.039	1.197	2.842
	I-RRIP	0.3822	1.0003	3.823	0.957	2.866
Average	LRU (baseline)		1.0000			
	SRRIP		1.0004			
	I-RRIP		1.0079			

Table 7: Task 2A: IPC, speedup over LRU, and L2 load misses per 1000 instructions.

Trace	L2 instr. MPKI	L2 data MPKI	L2C load MPKI	Speed
Trace1	−5.5%	+7.2%	−0.3%	−0.06%
Trace2	−35.3%	+7.3%	−22.0%	+2.41%
Trace3	−14.9%	+1.3%	−3.3%	+0.03%

Table 8: Change of each metric with I-RRIP relative to LRU (negative MPKI change = fewer misses).

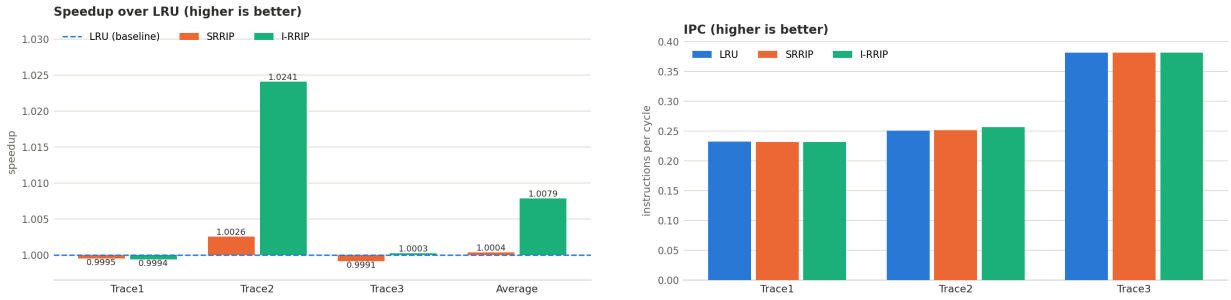


Figure 5: Task 2A: speedup over LRU (dashed line: LRU = 1.0) and IPC per trace.

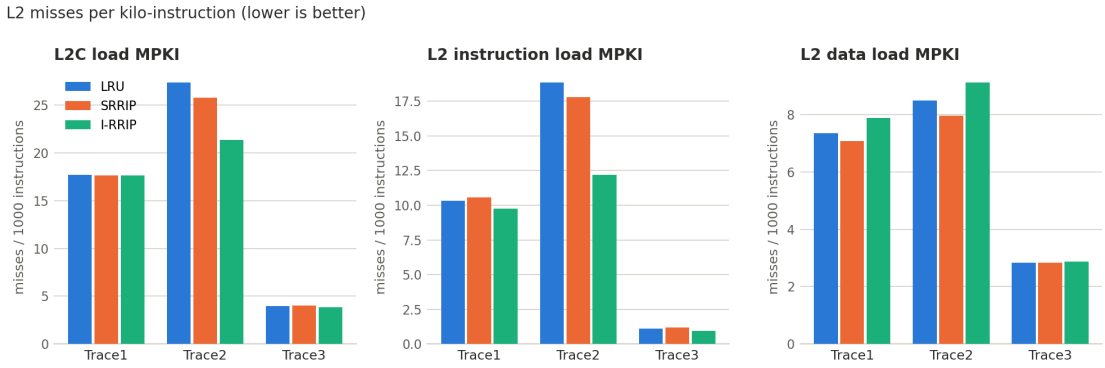


Figure 6: Task 2A: L2 misses per 1000 instructions: all loads, instruction loads, data loads.

Observations.

- **The goal is met.** I-RRIP lowers L2 instruction misses on every trace: −5.5% (Trace1), −35.3% (Trace2) and −14.9% (Trace3).
- **Trace2 gains the most.** It has many more code misses than data misses (18.85 vs. 8.50 MPKI with LRU). There I-RRIP removes 22% of all L2 load misses and runs 2.4% faster.
- **The price is a few more data misses**, because data lines are now evicted earlier. On Trace1 and Trace3 code misses were already low, so the small gain and the small loss cancel out and the speed stays the same (within 0.1%).
- **SRRIP alone hardly helps** (average speedup 1.0004). The improvement comes from treating code lines differently.

3.6 Design choices

We changed all three parts of the policy: insertion (RRPV of a new line), promotion (RRPV after a hit) and eviction (which line to evict). To check that each change is useful, we also ran the policy with only one change enabled at a time, using the same binary with different settings (Table 9, Figure 7).

Version	Avg. speedup	Trace2 speedup	Trace2 instr. MPKI	Trace2 data MPKI
LRU (baseline)	1.0000	1.0000	18.853	8.504
SRRIP	1.0004	1.0026	17.781	7.965
Only the two small fixes (§3.6)	1.0039	1.0111	15.619	7.873
+ promotion change only	1.0051	1.0149	14.478	8.376
+ insertion change only	1.0054	1.0155	14.711	8.363
+ eviction change only	1.0080	1.0236	12.653	8.428
All, but new code at RRPV 0	1.0072	1.0229	12.525	9.274
I-RRIP (all three)	**1.0079**	**1.0241**	**12.204**	9.125

Table 9: Each change on its own. All variants except LRU and SRRIP include the two small fixes.

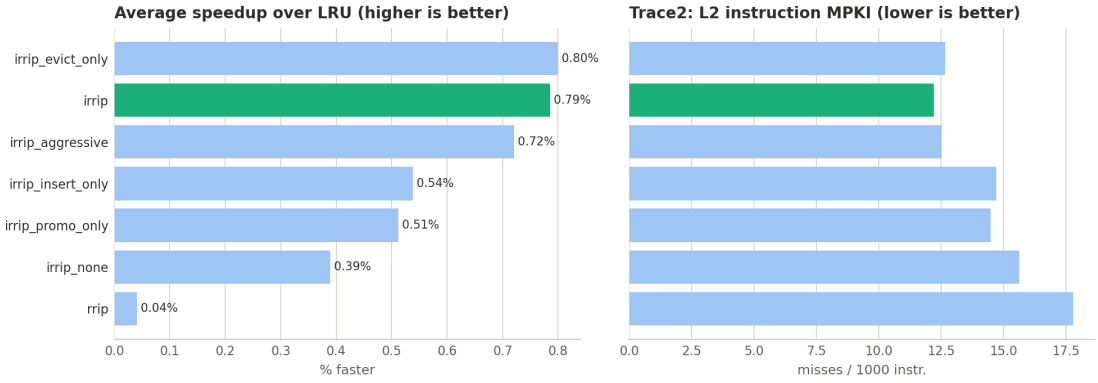


Figure 7: The comparison of Table 9 as a chart; I-RRIP is green. `irrip_none` = only the two small fixes; `irrip_promo_only` / `irrip_insert_only` / `irrip_evict_only` = one change only; `irrip_aggressive` = new code at RRPV 0.

Insertion: new code lines at RRPV 1, new data lines at RRPV 2. SRRIP inserts every new line at RRPV 2. A code line inserted at 1 gets one extra round before it can be evicted, so it has more time to be reused, while data is not hurt more than needed. We did not use RRPV 0 for code: then code that runs only once (e.g. start-up code) would be protected as much as code that is used again and again. Evidence: with RRPV 0, data misses go up (9.27 vs. 9.13 MPKI on Trace2), code misses do not go down, and the average speedup is lower (1.0072 vs. 1.0079). On its own, this change raises the Trace2 speedup from 1.0111 to 1.0155.

Promotion: a reused code line goes to RRPV 0, a reused data line only to RRPV 1. With SRRIP, popular data competes equally with popular code. Now code that is reused always ranks above data that is reused. A hit never makes a line's RRPV worse. On its own, this change raises the Trace2 speedup from 1.0111 to 1.0149 and lowers code misses from 15.62 to 14.48 MPKI.

Eviction: data before code. When a line must be evicted, we first look for a data line at RRPV 3; only if there is none do we take a code line at RRPV 3. If no line is at RRPV 3, all RRPVs are incremented and the search repeats. Among lines that are "not needed soon", losing a data line usually hurts less than losing a code line. This is the strongest single change: on its own it gives an average speedup of 1.0080 and cuts Trace2 code misses from 18.85 (LRU) to 12.65 MPKI.

Two small fixes (not about code vs. data).

- **Use an empty way first.** If a way in the set is empty, it is used instead of evicting a line. SRRIP's search did not check this.
- **Writebacks do not count as reuse.** A writeback is the L1D pushing a modified line down, not the program reading it again. SRRIP still set such lines to RRPV 0; we leave their RRPV unchanged (the repository's LRU policy does the same).

These two alone give an average speedup of 1.0039 (SRRIP: 1.0004). The three code-vs-data changes then take it to 1.0079.

Why this combination.

- It has the fewest L2 instruction misses on the code-heavy trace: 12.20 MPKI, −35.3% compared with LRU, which is exactly what Task 2A asks for.
- It is the fastest on Trace2 (1.0241) and its average speedup (1.0079) is within 0.01% of the best single change (eviction only, 1.0080).
- Each of the three changes helps on its own, and the one alternative we tried for insertion (RRPV 0) was worse.
- Limitation: on traces with few code misses (Trace1, Trace3) the trade-off brings no gain. Enabling it only when code matters is the idea of Task 2B.

3.7 Storage overhead

Stored state	Size (L2: 1024 sets × 8 ways = 8192 lines)
RRPV, 2 bits per line (same as SRRIP)	8192 × 2 bits = 2 KB
Instruction bit, 1 bit per line	8192 × 1 bit = 1 KB
Total	**3 KB**

Table 10: Storage needed by I-RRIP: 0.6% of the 512 KB cache and 1 KB more than SRRIP.

4 Task 2B: Dynamic I-RRIP (Set Dueling)

Policy file: `replacement/dyn_irrip.l2c_repl`.

4.1 In short

- I-RRIP (Task 2A) always protects instruction lines. That is good when code misses dominate, but in a phase where data matters more it evicts data early for nothing.
- Dynamic I-RRIP decides at run time. A few *leader* sets always run plain SRRIP and measure how much of the L2 traffic is instruction traffic. If that share is above a threshold, all other sets (*followers*) use I-RRIP; otherwise they use SRRIP.
- Result: 2.23% faster than LRU on Trace2 (code-heavy) and 0.68% faster on average, with 33% fewer L2 instruction misses on Trace2. It keeps about 87% of the I-RRIP gain while being able to step back to SRRIP when code stops mattering.

4.2 Design

Leader and follower sets. Of the 1024 L2 sets, 32 are leader sets (about 3% of the cache). Leader sets always use plain SRRIP: insert at RRPV 2, set RRPV 0 on a hit, evict the first line at RRPV 3. The other 992 sets are followers and use whichever mode the leaders last chose: I-RRIP (exactly the Task 2A rules) or SRRIP. The leaders are spread over the whole cache, one in every block of 32 sets, at a position that moves by one in each block (sets 0, 33, 66, ..., 1023), so they do not all share the same low index bits.

What the leaders count. In a leader set, every L2 access that is not a writeback updates four counters: instruction accesses, instruction misses, total accesses and total misses. A hit counts as one access; a fill (i.e. a miss) counts as one access and one miss. Instruction lines are recognised with the `isInstruction` input added in Task 2A. Writebacks are skipped because they are the L1D evicting dirty lines, not the program asking for anything.

The decision. Every 1024 leader accesses (one *window*) the policy computes the ratio from the assignment:

$$\text{ratio} = \frac{\text{instr. accesses} + \text{instr. misses}}{\text{total accesses} + \text{total misses}}$$

If ratio > threshold (default 0.25), followers use I-RRIP; otherwise SRRIP. The ratio is the share of L2 work caused by code, with misses counted twice because a miss costs much more than a hit. After each decision the counters are halved, so they keep a memory of about the last two windows.

Follower sets. A follower in I-RRIP mode applies all three Task 2A rules (insertion, promotion including the writeback rule, and data-first eviction). In SRRIP mode it behaves exactly like a leader. Both modes use an empty way first. Because the RRPV bits mean the same in both modes, switching needs no reset: lines keep their RRPVs and the new rules apply from the next access.

4.3 Experimental setup

- Same setup as Task 2A: 25M warmup + 25M measured instructions per trace, no L1D prefetcher, LRU in every other cache, baseline = LRU at the L2. Speedup = IPC / IPC(LRU); averages are geometric means.

- All settings (threshold, number of leaders, window) are read from environment variables (`DYN_THRESHOLD`, `DYN_LEADERS`, `DYN_WINDOW`), so the sensitivity runs use one binary and differ only in the setting being studied. Runner: `scripts/run_task2b.sh`; tables and plots: `scripts/parse_results.py`, `plot_results.py`, `plot_task2b.py`.
- The policy prints its own statistics after warmup: number of decisions, mode switches, average ratio and the share of follower accesses served in I-RRIP mode.

4.4 Results (threshold 0.25, 32 leader sets)

Trace	Policy	IPC	Speedup	L2C load MPKI	L2 instr. MPKI	L2 data MPKI
Trace1	LRU (baseline)	0.2323	1.0000	17.696	10.338	7.358
	SRRIP	0.2322	0.9995	17.639	10.562	7.077
	I-RRIP (Task 2A)	0.2322	0.9994	17.649	9.765	7.884
	Dyn-I-RRIP	0.2321	0.9991	17.651	9.788	7.864
Trace2	LRU (baseline)	0.2512	1.0000	27.357	18.853	8.504
	SRRIP	0.2518	1.0026	25.746	17.781	7.965
	I-RRIP (Task 2A)	0.2572	1.0241	21.329	12.204	9.125
	Dyn-I-RRIP	0.2568	1.0223	21.724	12.631	9.093
Trace3	LRU (baseline)	0.3821	1.0000	3.954	1.124	2.830
	SRRIP	0.3817	0.9991	4.039	1.197	2.842
	I-RRIP (Task 2A)	0.3822	1.0003	3.823	0.957	2.866
	Dyn-I-RRIP	0.3818	0.9993	3.931	1.064	2.868
Average	LRU (baseline)		1.0000			
	SRRIP		1.0004			
	I-RRIP (Task 2A)		1.0079			
	Dyn-I-RRIP		1.0068			

Table 11: Task 2B: IPC, speedup over LRU, and L2 load misses per 1000 instructions.

Trace	L2 instr. MPKI	L2 data MPKI	L2C load MPKI	Speed vs. LRU	Speed vs. I-RRIP
Trace1	−5.3%	+6.9%	−0.2%	−0.09%	−0.03%
Trace2	−33.0%	+6.9%	−20.6%	+2.23%	−0.18%
Trace3	−5.4%	+1.3%	−0.6%	−0.07%	−0.10%
Average				+0.68%	−0.10%

Table 12: Change of each metric with Dyn-I-RRIP relative to LRU, and its speed relative to static I-RRIP.

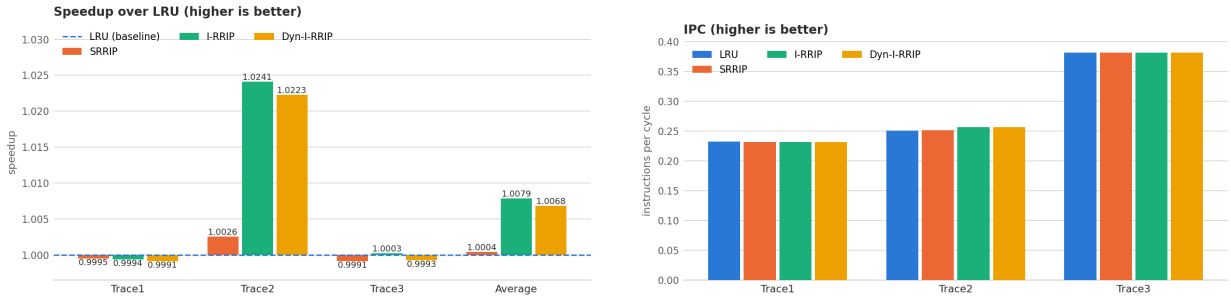


Figure 8: Task 2B: speedup over LRU (dashed line: LRU = 1.0) and IPC per trace.

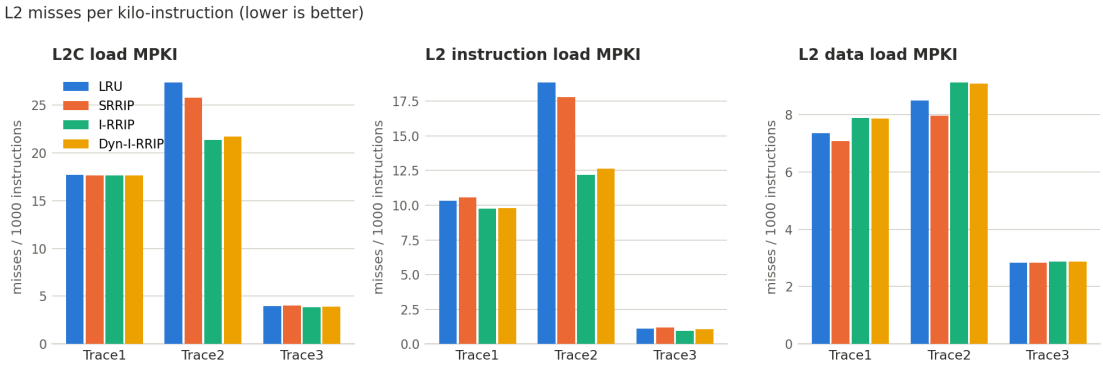


Figure 9: Task 2B: L2 misses per 1000 instructions: all loads, instruction loads, data loads.

Observations.

- **Dyn-I-RRIP still meets the Task 2A goal.** L2 instruction misses fall on every trace compared with LRU: −5.3% (Trace1), −33.0% (Trace2), −5.4% (Trace3).
- **Trace2 is where it pays off:** 20.6% fewer L2 load misses and 2.23% faster than LRU. The leaders measure an average ratio of 0.60 there, far above the threshold, so 94% of follower accesses are served in I-RRIP mode.
- **Trace1 and Trace3 stay within 0.1% of LRU**, as with I-RRIP. Code and data effects cancel on these traces, so there is little to win or lose.
- **Against static I-RRIP** the dynamic policy is 0.10% slower on average (0.18% on Trace2); Section 4.7 explains why.

4.5 Sensitivity study

We varied the decision threshold over five values around the suggested 0.25 (32 leaders), and then the number of leader sets (16 / 32 / 64, threshold 0.25). Everything else was unchanged.

Thresh.	Trace	IPC	Speedup	L2C load MPKI	L2 instr. MPKI	L2 data MPKI	% in I-RRIP mode	Switches
0.10	Trace1	0.2322	0.9995	17.649	9.789	7.860	100.0	0
	Trace2	0.2571	1.0235	21.453	12.349	9.104	100.0	0
	Trace3	0.3821	1.0002	3.831	0.966	2.865	100.0	0
	Average		1.0077					
0.20	Trace1	0.2321	0.9991	17.651	9.788	7.864	100.0	0
	Trace2	0.2568	1.0223	21.724	12.629	9.095	95.8	2
	Trace3	0.3819	0.9996	3.914	1.052	2.863	71.9	1
	Average		1.0069					
0.25	Trace1	0.2321	0.9991	17.651	9.788	7.864	100.0	0
	Trace2	0.2568	1.0223	21.724	12.631	9.093	93.7	2
	Trace3	0.3818	0.9993	3.931	1.064	2.868	57.5	1
	Average		1.0068					
0.30	Trace1	0.2322	0.9992	17.680	9.810	7.870	100.0	0
	Trace2	0.2567	1.0222	21.731	12.637	9.094	93.7	2
	Trace3	0.3817	0.9991	3.998	1.121	2.878	42.3	1
	Average		1.0068					
0.40	Trace1	0.2322	0.9993	17.604	9.841	7.762	88.4	4
	Trace2	0.2565	1.0214	21.892	12.820	9.072	91.5	2
	Trace3	0.3817	0.9991	3.998	1.119	2.878	42.5	1
	Average		1.0065					

Table 13: Threshold sensitivity (32 leader sets). "% in I-RRIP mode" = share of follower accesses served in I-RRIP mode; "Switches" = mode changes after warmup.

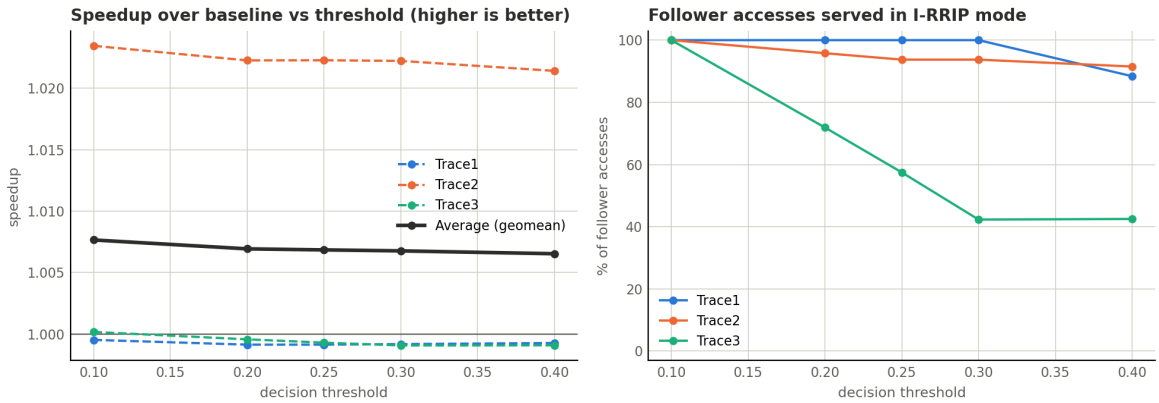


Figure 10: Left: speedup vs. threshold. Right: share of time the followers spend in I-RRIP mode.

Leader sets	% of cache	Trace1	Trace2	Trace3	Avg. speedup	Trace2 instr. MPKI
16	1.6%	0.9994	1.0219	0.9997	1.0069	12.691
32	3.1%	0.9991	1.0223	0.9993	1.0068	12.631
64	6.3%	0.9994	1.0222	0.9991	1.0068	12.693

Table 14: Leader-set sensitivity (threshold 0.25). Speedups over LRU.

What the sensitivity study shows.

- **The result is flat.** Over a 4× range of thresholds the average speedup only moves between 1.0065 and 1.0077 (0.12% spread); Trace2 stays between 1.0214 and 1.0235. A wrong threshold costs little.
- **Why it is flat:** the leaders measure an average ratio of about 0.44 on Trace1 and 0.60 on Trace2, above every threshold we tried. Only Trace3 (average ratio 0.31) sits close to the thresholds, and its time in I-RRIP mode falls from 100% to 42% as the threshold rises (Figure 10, right). Since I-RRIP and SRRIP differ by only 0.1% on Trace3, this barely shows in the speedup.
- **0.10 is the fastest (1.0077) but not really dynamic:** the ratio never drops that low, so the followers stay in I-RRIP mode 100% of the time and the policy becomes static I-RRIP minus the 3% of sets that the leaders keep in SRRIP.
- **0.40 starts to switch on Trace1** (4 switches, 88% in I-RRIP). There data misses go down while code misses rise only slightly, but Trace2 is pushed out of I-RRIP mode more often and loses more, so the average is the lowest of the five.
- **We kept 0.25.** Among thresholds where switching actually happens, 0.20, 0.25 and 0.30 are equal within 0.02%. 0.25 sits in the middle of that flat region, so it is the least sensitive choice, and it still lets the policy leave I-RRIP mode in a phase where code becomes a small part of the L2 traffic.

4.6 Design choices

Number of leader sets: 32.

- Too few leaders means a noisy sample: with 1024 accesses per window spread over only a few sets, a short burst in one set can decide the mode for the whole cache. More leaders also give more decisions per run (Trace2: 25 decisions with 16 leaders, 46 with 32, 97 with 64), so the policy reacts faster to a new phase.
- Too many leaders costs performance, because leader sets always run SRRIP and never get the I-RRIP benefit. With 64 leaders, 6.3% of the cache is permanently in SRRIP mode.
- Measured: 16, 32 and 64 leaders give average speedups of 1.0069, 1.0068 and 1.0068 (Table 14), the same within 0.01%. With equal performance we chose the middle value: 32 leaders (3.1% of sets) is the size commonly used for set dueling in a 1024-set cache (as in DIP/DRRIP), gives a sample twice as large as 16 for a less noisy decision, and loses half as many sets to SRRIP as 64.

Decision logic.

- **Leaders run SRRIP, not I-RRIP.** The policy asks "how much does code matter if we don't protect it?". Leaders running I-RRIP would hide code misses, the ratio would drop, and the policy would switch protection off exactly because protection was working.
- **Misses count twice in the ratio** (once as an access, once as a miss). A workload whose code mostly hits in the L2 has a lower ratio than one whose code keeps missing, which is what we want: protection only helps when code is being lost.

- **Window of 1024 leader accesses + halving.** A fixed window gives regular decisions; halving keeps a memory of about the last two windows. During the 25M measured instructions this gave 26 / 46 / 6 decisions and only 0 / 2 / 1 switches on Trace1 / 2 / 3: the policy follows real phase changes but does not thrash.
- **Threshold 0.25** from the assignment hint, kept after the sensitivity study.

4.7 Why Dynamic I-RRIP does not beat static I-RRIP on these traces

This is expected and does not mean the dynamic policy is wrong. Two things cost it a little:

- **The leaders never use I-RRIP.** 3.1% of the cache always runs SRRIP, so on a trace where I-RRIP is always the right choice (Trace1, Trace2) the dynamic policy gives up the I-RRIP gain in those sets. This is the fixed price of set dueling: Trace2 runs in I-RRIP mode for 94% of follower accesses and still loses 0.18%.
- **Switching only helps if the other mode is better.** When Trace3 drops into SRRIP mode (43% of the time) it falls from I-RRIP's 1.0003 to 0.9993, because on Trace3 SRRIP is slightly worse than I-RRIP (Task 2A: 0.9991 vs. 1.0003).

The deeper reason is that on these three traces static I-RRIP never really loses: its worst result is −0.06% (Trace1). A dynamic policy pays the small set-dueling cost to protect against a case that does not occur here. It would win on a workload with a long data-heavy phase where code fits in the L1I, where static I-RRIP would raise data misses for nothing (on Trace1 static I-RRIP already raises data misses by 7%). The ratio also measures how much code the L2 sees rather than directly comparing the misses of the two policies. A natural next step would be a second group of leaders running I-RRIP and comparing the misses of both groups; we kept the single-group design from the hint because it is cheaper and its loss here is only 0.10%.

4.8 Storage overhead

Stored state	Size (L2: 1024 sets × 8 ways = 8192 lines)
RRPV, 2 bits per line (same as SRRIP / I-RRIP)	8192×2 bits = 2 KB
Instruction bit, 1 bit per line (same as I-RRIP)	8192×1 bit = 1 KB
Leader-set flag	0 bits: computed from the set index (two 5-bit fields compared)
4 counters: instruction/total accesses and misses	4×12 bits = 48 bits (halving keeps each value ≤ 2047)
Window counter (0–1024)	11 bits
Follower mode bit (I-RRIP / SRRIP)	1 bit
Total	**3 KB + 60 bits** (under 8 bytes more than I-RRIP; about 0.6% of the 512 KB cache)

Table 15: Storage needed by Dynamic I-RRIP.

5 Task 3: SPP with Prefetch-Aware Dynamic I-RRIP

Policy file: `replacement/dyn_irrip.l2c_repl` (SPP at the L1D, `prefetcher/spp.l1d_pref`, unchanged from Task 1B).

5.1 Design

SPP issues its prefetches with `FILL_L1`. A prefetch that misses in the L2 is fetched from the LLC/DRAM and filled into *both* the L2 and the L1D. The L2 replacement policy sees these requests with `type == PREFETCH`. Task 3 adds one rule set for them to Dynamic I-RRIP. It is applied in every set (leader and follower) and in both follower modes, before any other rule:

Event	Dynamic I-RRIP (Task 2B)	Prefetch-aware Dynamic I-RRIP
Prefetch fill (miss in L2)	inserted like a data line (RRPV 2)	RRPV 3
Prefetch hit in L2	promoted like a data hit (RRPV 0 or 1)	RRPV unchanged
Prefetch in a leader set	counted in the dueling ratio	not counted
Demand hit on a prefetched line	–	promoted as usual (code 0 / data 1 or 0)

Table 16: Prefetch handling added to Dynamic I-RRIP.

The rules only use the request type that the cache already passes to `update_replacement_state()`, so they need **no extra storage**. Nothing needs to remember that a line was prefetched; once a demand load hits it, it is treated like any other line.

5.2 Rationale for handling prefetch packets

The L2 copy of a prefetched line is rarely used. The L1D copy serves the program; the L2 copy is only read if the line is evicted from the L1D and then requested again while still in the L2. With SPP and LRU at the L2, only 7.4%, 15.1% and 5.0% of the lines that SPP filled into the L2 were ever hit by a demand load (Table 19). Under Dynamic I-RRIP such a line would be inserted at RRPV 2 like demand data and, on a hit by a later prefetch, promoted like reused data. It would then occupy an L2 way for a long time and push out instruction lines, which is exactly what I-RRIP tries to prevent.

Insertion at RRPV 3. A prefetched line is a data line already at the maximum RRPV, so it is the first victim of the data-first eviction rule in I-RRIP mode and of the first-at-RRPV-3 rule in SRRIP mode. It is evicted before any instruction line and before demand data that has not aged yet. If the program does read it from the L2, the demand hit promotes it normally, so a useful prefetch is not lost. The measured cost is small: the share of L2 prefetch fills used by a demand load falls to 1.0%, 3.2% and 1.2%, i.e. about 1300, 250 and 350 fewer L2 hits per 25M instructions (at most 0.05 MPKI).

No promotion on a prefetch hit. A prefetch that hits in the L2 means the L1D is about to receive a copy of the line. That is not a use by the program, and after it the program reads the line from the L1D, not from the L2. Promoting the L2 copy would protect it exactly while it is least needed. Only demand hits (and, as in Task 2A, not writebacks) count as reuse.

Prefetches are not counted by the leader sets. The dueling ratio measures how much of the program's L2 demand traffic is instruction traffic. Every SPP prefetch is a data request (`L2C INSTRUCTION PREFETCH MPKI` is 0), and most of them miss in the L2, so counting them would add to the denominator twice and lower the ratio just because the prefetcher is active. On Trace3, whose ratio (0.35) is the closest to the 0.25 threshold, the 9910 prefetches (9590 misses) would add about 9% to the denominator (131k L2 load accesses + 91k misses) and pull the ratio towards the threshold. With prefetches excluded, the leaders measure nearly the same ratio as Dynamic I-RRIP without a prefetcher (Table 20), so the mode decision keeps reflecting code demand. Writebacks are excluded for the same reason.

5.3 Experimental setup

Four configurations are compared on each trace:

- **No prefetching (LRU)**: the Task 1B baseline; all speedups are relative to this run.
- **SPP (LRU)**: SPP at the L1D, LRU at the L2 (Task 1B).
- **Dyn-I-RRIP**: no prefetcher, Dynamic I-RRIP at the L2 (Task 2B).
- **SPP + PF-aware Dyn-I-RRIP**: SPP at the L1D, prefetch-aware Dynamic I-RRIP at the L2 (Task 3; threshold 0.25, 32 leaders, window 1024).

Every other cache uses LRU and no other prefetcher is active. Tables and plots are produced by `scripts/task3_metrics.py` (`results/task3/task3_metrics.csv`, `task3_table.md`, `plots/`).

5.4 Results

	IPC				Speedup over no prefetching (LRU)		
Trace	No PF	SPP	Dyn-I-RRIP	SPP + PF-aware	SPP	Dyn-I-RRIP	SPP + PF-aware
Trace1	0.2322	**0.2332**	0.2321	0.2327	**1.0042**	0.9997	1.0021
Trace2	0.2511	0.2513	**0.2568**	0.2552	1.0008	**1.0224**	1.0163
Trace3	0.3821	0.3829	0.3818	**0.3830**	1.0023	0.9993	**1.0024**
Average	0.2885	0.2891	0.2902	**0.2903**	1.0024	**1.0071**	1.0069

Table 17: Task 3: IPC and speedup (Average: arithmetic mean for IPC, geometric mean for speedup). Best value per row in bold.

Trace	Configuration	L1D load MPKI	L2C load MPKI	L2 instr. MPKI	L2 data MPKI
Trace1	No prefetching (LRU)	10.679	17.730	10.384	7.346
	SPP (LRU)	10.062	17.214	10.389	6.825
	Dyn-I-RRIP	10.680	17.651	9.788	7.864
	SPP + PF-aware Dyn-I-RRIP	10.072	17.172	9.848	7.323
Trace2	No prefetching (LRU)	13.534	27.923	19.391	8.533
	SPP (LRU)	13.446	27.827	19.326	8.500
	Dyn-I-RRIP	13.541	21.724	12.631	9.093
	SPP + PF-aware Dyn-I-RRIP	13.457	23.206	14.218	8.988
Trace3	No prefetching (LRU)	3.403	3.959	1.126	2.833
	SPP (LRU)	3.285	3.703	1.129	2.574
	Dyn-I-RRIP	3.403	3.931	1.064	2.868
	SPP + PF-aware Dyn-I-RRIP	3.277	3.649	1.053	2.595
Average	No prefetching (LRU)	9.205	16.538	10.300	6.237
	SPP (LRU)	8.931	16.248	10.281	5.967
	Dyn-I-RRIP	9.208	14.436	7.827	6.608
	SPP + PF-aware Dyn-I-RRIP	8.935	14.675	8.373	6.302

Table 18: Task 3: load MPKI of the L1D, the L2C, and L2 instruction and data loads.

		L1D				L2 copies of prefetched lines		
Trace	Configuration	Accuracy	Coverage	PF issued	PF useful	Fills	Demand hits	% used
Trace1	SPP (LRU)	60.72%	5.78%	28454	17278	20484	1513	7.39
	SPP + PF-aware	60.94%	5.69%	27752	16913	21985	221	1.01
Trace2	SPP (LRU)	57.47%	0.65%	5211	2995	2279	345	15.14
	SPP + PF-aware	56.67%	0.57%	5394	3057	2890	93	3.22
Trace3	SPP (LRU)	33.36%	3.48%	9496	3168	9112	455	4.99
	SPP + PF-aware	33.73%	3.70%	9910	3343	9590	110	1.15
Average	SPP (LRU)	50.52%	3.30%					
	SPP + PF-aware	50.45%	3.32%					

Table 19: Task 3: L1D prefetch accuracy and coverage, and what happens to the L2 copies of prefetched lines (fills = prefetches that missed in the L2; demand hits = those L2 lines later hit by a demand load). "SPP + PF-aware" = SPP + prefetch-aware Dyn-I-RRIP.

	Average ratio		% in I-RRIP mode		Mode switches	
Trace	Dyn	SPP + Dyn	Dyn	SPP + Dyn	Dyn	SPP + Dyn
Trace1	0.439	0.445	100.0	100.0	0	0
Trace2	0.600	0.634	93.7	95.8	2	2
Trace3	0.308	0.350	57.5	63.8	1	1

Table 20: Set-dueling decision of Dynamic I-RRIP without a prefetcher (Dyn) and with SPP and the prefetch rules (SPP + Dyn). "% in I-RRIP mode" = share of follower accesses served in I-RRIP mode.

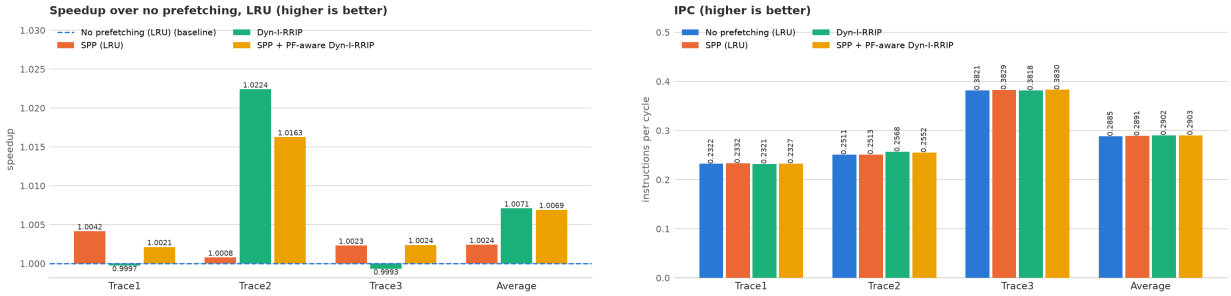


Figure 11: Task 3: speedup over no prefetching with LRU (dashed line) and IPC.

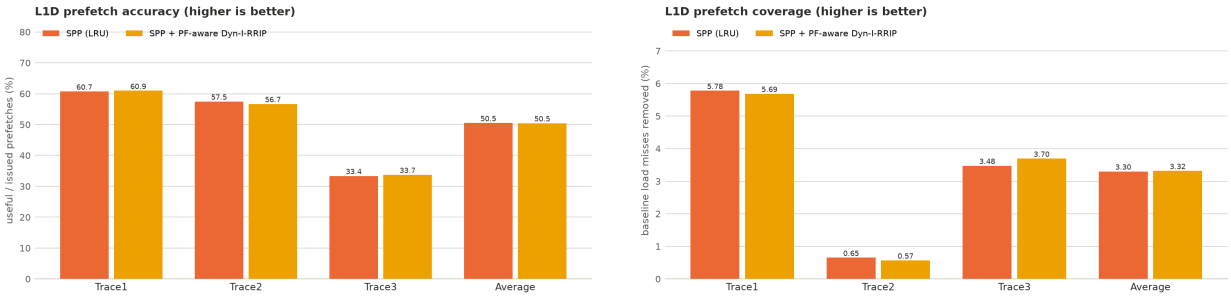


Figure 12: Task 3: L1D prefetch accuracy and coverage of SPP with LRU and with prefetch-aware Dynamic I-RRIP at the L2.

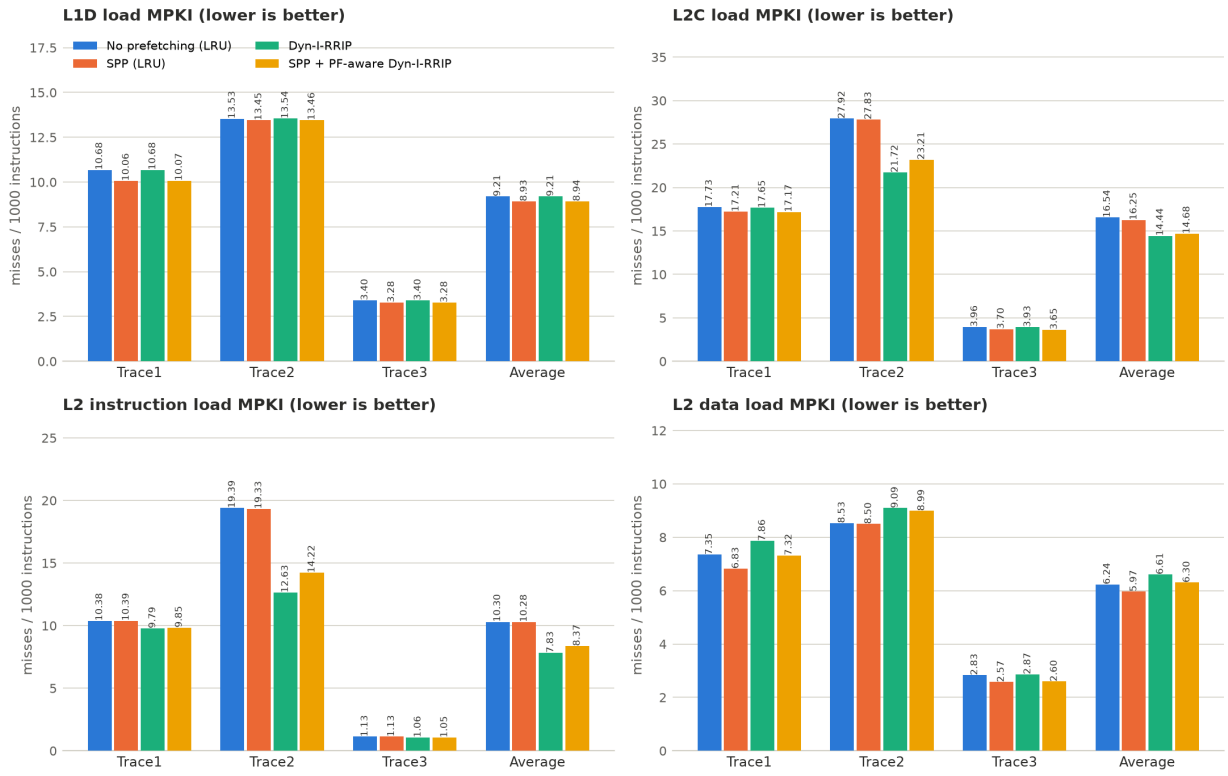


Figure 13: Task 3: L1D, L2C, L2 instruction and L2 data load MPKI.

5.5 Observations

- **The two mechanisms combine on the misses they target.** The combination keeps SPP's L1D benefit (L1D load MPKI 8.935 vs. 8.931 for SPP alone; same accuracy and coverage within 1%) and most of Dynamic I-RRIP's L2 instruction benefit (average L2 instruction MPKI 8.373 vs. 10.281 for SPP with LRU). It also has the lowest L2C load MPKI on Trace1 and Trace3.

- **SPP behaves the same under the new L2 policy.** L1D accuracy and coverage remain almost unchanged (Table 19), so inserting prefetched lines at RRPV 3 in the L2 does not seem to hurt the prefetcher: what matters to it is the L1D copy.

- **IPC.** The combination has the highest average IPC (0.2903 vs. 0.2902 for Dyn-I-RRIP and 0.2891 for SPP) and is the best configuration on Trace3 (1.0024). Its geometric-mean speedup (1.0069) is equal to Dynamic I-RRIP's (1.0071) within noise and well above SPP's (1.0024). It does *not* beat both standalone mechanisms on every trace: on Trace1 it is between them (1.0021 vs. SPP's 1.0042), and on Trace2 it is 0.6% behind Dynamic I-RRIP (1.0163 vs. 1.0224).